

An adaptive method for identifying marine areas of high conservation priority

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Abstract: Identifying priority areas for biodiversity conservation is particularly challenging in the marine environment due to the open and dynamic nature of the ocean, the paucity of information on species distribution, and the necessary balance between marine biodiversity conservation and essential supporting services such as seafood provision. We used the Patagonian seabird breeding community as a case study to propose an integrated and adaptive method for delimiting key marine areas for conservation. Priority areas were defined through a free decision-support tool (Marxan) that included projected at-sea distributions of seabirds (approximately 2,225,000 individuals of 14 species); BirdLife Important Bird and Biodiversity Areas (IBAs) for pelagic bird species; and the economic costs of potential regulations in fishing practices. The proposed reserve network encompassed approximately 300,000 km² that was largely concentrated in northern and southern inshore and northern and central offshore regions. This reserve network exceeded the minimum threshold of 20% conservation of the abundance of each species proposed by the World Parks Congress. Based on marine currents in the study area, we further identified the 3 primary water masses that may influence areas of conservation priority through water inflow. Our reserve network may benefit from enhanced marine productivity in these highly connected areas, but they may be threatened by human impacts such as marine pollution. Our method of reserve network design is an important advance with respect to the more classical approaches based on criteria defined for one or a few species and may be particularly useful when information on spatial patterns is data deficient. Our approach also accommodates addition of new information on seabird distribution and population dynamics, human activities, and alterations in the marine environment.

Keywords: marine protected areas, Marxan, ocean currents, remote sensing, reserve design, seabird at-sea distribution

Un Método Adaptivo para Identificar las Áreas Marinas de Alta Prioridad para la Conservación

Resumen: La identificación de áreas prioritarias para la conservación de la biodiversidad es un reto particular para los ambientes marinos debido a la naturaleza abierta y dinámica del océano, la escasez de información sobre la distribución de especies, y el balance necesario entre la conservación de la biodiversidad marina y los servicios esenciales que nos proporciona como la provisión de alimentos de origen marino. Utilizamos a la comunidad reproductora de aves marinas de la Patagonia como caso de estudio para proponer un método integrado y adaptativo para delimitar áreas marinas clave para la conservación. Las áreas prioritarias estuvieron definidas por medio de una herramienta gratuita de apoyo a las decisiones (Marxan) que incluyó las distribuciones oceánicas proyectadas de las aves marinas (aproximadamente 2, 225, 000 individuos de 14 especies); Áreas Importantes de Aves y Biodiversidad BirdLife (IBAs) para las especies pelágicas; y los costos económicos de las regulaciones potenciales en las prácticas pesqueras. La

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red de reservas propuesta englobó aproximadamente 300, 000 km² que estuvo concentrada en su mayoría en las regiones costeras del norte y del sur, y en las zonas centro y norte en el límite de la plataforma continental. Con esta red de reservas se superó el umbral mínimo del 20% de conservación de la abundancia de cada especie propuesto por el Congreso Mundial de Parques. A partir de las corrientes marinas prioritarias en el área de estudio identificamos las tres masas principales de agua que pueden influir sobre las áreas prioritarias para la conservación a través del flujo de agua proveniente de otras zonas externas a las de conservación. Estas aguas externas pueden modificar la calidad de las aguas de la red de reservas propuestas, beneficiándolas con un aumento de la productividad pero también amenazándolas por el riesgo de que las alcance aguas contaminadas por impactos humanos. Nuestra metodología de diseño de red de reservas es un avance importante con respecto a las estrategias más clásicas basadas en criterios definidos para una o pocas especies y puede ser útil particularmente cuando la información sobre los patrones espaciales de las especies a conservar es deficiente. Nuestra estrategia es adaptativa a la integración de nueva información sobre la distribución de aves marinas y sus dinámicas poblacionales, actividades humanas, y las alteraciones del ambiente marino.

Palabras Clave: áreas marinas protegidas, corrientes oceánicas, diseño de reservas, distribución oceánica de aves marinas, Marxan, telemetría

摘要: 确定海洋环境生物多样性保护的优先区域极具挑战, 这是因为海洋具有开放和动态变化的特性, 且物种分布信息缺乏, 还必须平衡海洋生物多样性保护和基础支持服务 (如提供海产品)。我们以巴塔哥尼亚的海鸟繁殖群作为案例研究, 提出了一个综合的适应性方法来划定海洋保护的关键区域。我们用免费的决策支持工具 (Marxan软件) 来确定优先保护区域, 考虑了海鸟在海上的估计分布 (14个物种, 约2,225,000只个体)、国际鸟盟 (BirdLife) 《重要鸟类及生物多样性区域》中的远洋鸟种, 以及对捕鱼活动的监管可能带来的经济成本。我们得到的保护区网络约 300,000 平方公里, 主要集中在近海北部和南部、远海北部和中部地区。这个保护区网络超出了世界公园会议 (World Parks Congress) 提出的保护每个物种丰度20%的最低限度。根据研究地点的洋流, 我们进一步确定了三个通过海水流入影响优先保护区域的水团。我们的保护区网络可以得益于这些高度连接的地区海洋生产力的提高, 但这些地区也可能受人类活动的威胁, 如海洋污染。相比于以一个或几个物种为标准进行保护区网络设计的传统方法, 我们的方法取得了重要的进步, 特别是可以用于空间格局数据缺乏的情况。另外, 我们的方法还纳入了海鸟分布及种群动态、人类活动和海洋环境变化的新信息。【翻译: 胡怡思; 审校: 魏辅文】

关键词: 海洋保护地, Marxan 软件, 洋流, 遥感, 海鸟的海上分布, 保护区设计

Introduction

Human activities have extensively affected the marine environment and caused loss of marine biodiversity and changes to the structure and functioning of marine ecosystems globally (Lewison et al. 2004; Cury et al. 2011). Consequently, marine conservation is of social, political, and scientific concern worldwide. The conservation of marine biodiversity is, however, particularly challenging because it requires delimiting meaningful protected marine areas in a large, remote, and continuous system (Pérez-Jorge et al. 2015); considering the balance between marine biodiversity conservation and fisheries (Agardy et al. 2003; Bennett & Dearden 2014); and incorporating dynamic processes that affect physical and biological oceanographic features (Game et al. 2009; Grantham et al. 2011).

As flagship species, seabirds have been used to leverage support for marine biodiversity conservation and as surrogates of marine biodiversity (Furness & Camphuysen 1997; Grémillet & Charmantier 2010). Their life-history traits and population parameters are a reflection of the dynamic processes within food webs and are determined by the distribution of species in lower trophic levels (Frederiksen et al. 2006; Ramírez et al. 2014; Sandvik

et al. 2016). The delimitation of effective marine reserves for the conservation of marine biodiversity in general, and seabirds in particular, depends on an accurate understanding of the spatial distribution and abundance of foraging seabirds (Karpouzi et al. 2007; Votier et al. 2010; Cury et al. 2011). The recent miniaturization of tracking devices has provided the deepest insights thus far into the at-sea distribution of seabirds (Ropert-Coudert & Wilson 2005). However, this information is largely restricted to particular species and colonies (Raymond et al. 2014), which hampers analyses of the spatial patterns of the at-sea distribution of seabird communities. As central-place foragers, the at-sea distributions of breeding seabirds are constrained by their need to return to breeding places to incubate or feed their offspring (Weimerskirch 2007). Thus, conducting colony-based counts and projecting at-sea distributions based on known foraging ranges provides a simple, inexpensive, and quick way to predict at-sea distribution of breeding seabirds (e.g., Grecian et al. 2012; Soanes et al. 2016), although this method lacks the rigor of biologging. Moreover, the method can be applied to entire seabird communities over large areas and accounts for the high-energy demands of reproduction, which is a large factor in population dynamics (Grecian et al. 2012).

Fisheries pose a threat to marine biodiversity (Halpern et al. 2008), and marine conservation policies would be ineffective if this major component of the marine environment were ignored (Lewison et al. 2004; Bennett & Dearden 2014; Scales et al. 2014). However, few approaches explicitly include potential economic costs from fishing regulations in reserve-network design (but see Allnutt et al. 2012; Mazar et al. 2014). In part, this is because of the lack of reliable, spatially explicit information on the distribution of fishing vessels. Fortunately, recent modeling advances, based on automatic identification systems (AIS) tracking data for fishing vessels, are providing reliable information on the spatial distribution of fishing effort for inclusion in systematic conservation planning (Kroodsma et al. 2018).

Identifying priority conservation areas is particularly challenging in the marine environment due to the open and dynamic nature of the ocean, where water masses are connected through marine currents (Afán et al. 2015). Thus, environmental processes and human impacts may be influencing conservation targets far beyond the limits or borders of priority conservation areas. Satellite remote-sensing data and recent advances in oceanographic models now provide the means to account for the dynamism of ocean systems so that marine areas that may affect priority conservation areas through water inflows can be identified.

We devised a new approach to identify key areas for seabird conservation in one of the most productive systems in the world, the Patagonian Sea (Falabella et al. 2009). We combined projected at-sea distributions of seabirds breeding on the Argentinean coast and Birdlife International Important Bird and Biodiversity Areas (IBAs), delimited for pelagic species, with spatially explicit information on fishing. We delimited high-priority areas for seabird conservation that fulfill species-specific conservation targets (based on species' conservation status) while minimizing the economic costs of potential fishing regulations. Based on marine currents prevalent in the study area, we also identified water masses that may influence areas of conservation priority through water inflow and that are susceptible to external threats such as marine pollution (Ruoppolo et al. 2012; Marinho et al. 2013) or biomass drift (Afán et al. 2015). We sought to develop a multidisciplinary and adaptive framework that would balance marine biodiversity conservation and ecosystem services.

Methods

Projected Distribution of Seabirds

We focused on the seabird community (14 species, Table 1) breeding on the coast of Argentinean Patagonia from Península Valdés to Tierra del Fuego (41.5–55.5°S) (Fig. 1a). By including BirdLife International marine

IBAs (<https://maps.birdlife.org/marineIBAs/default.html>) in our reserve network design, we could also consider the distribution of pelagic seabirds feeding in Argentinean waters that breed in nearby areas (e.g., albatrosses, petrels, and shearwaters from the Falkland-Malvinas Islands) and northern species that use this region as wintering quarters.

The at-sea distribution of breeding seabirds was projected based on data from colony censuses carried out from 1993 to 1995 (Yorio et al. 1998) and information on central-place foraging distributions derived from previous tracking studies along the Patagonian coast on Magellanic Penguin (*Spheniscus magellanicus*), South American Tern (*Sterna hirundinacea*), Southern Giant Petrel (*Macronectes giganteus*), Olrog's Gull (*Larus atlanticus*), and King Cormorant (*Phalacrocorax atriceps*). We used these data to delimit maximum foraging ranges of inshore (60 km) and offshore species (500 km) (Bugoni & Vooren 2005; Wilson et al. 2005; Boersma et al. 2009; Copello et al. 2011; Suárez et al. 2012; Wilson et al. 2012).

We used a modified version of Hanski's (1998) isolation function to project the at-sea distribution of breeding species, $F_i = \sum \exp(-d_{ij} \cdot B_j) \cdot P_j$, where d_{ij} is the distance from each grid cell i to the colony j and B_j is the inverse of the Euclidean distance from each colony to the maximum foraging range of a given species (details given in Supporting Information). This resulted in a distance-dependent distribution of occurrence probabilities that we transformed into number of individuals by considering population size at colony j (P_j). The total number of individuals that potentially occur at a given distance according to this function was distributed uniformly among available sea pixels at each specific distance from the colony (Hanski 1998; Afán et al. 2014; Ramírez et al. 2014). Estimates of the total number of individuals per breeding species on a pixel basis were the sum of colony-specific at-sea projections (Supporting Information).

Fishing Costs

We included fisheries in our reserve-network design so that the economic costs associated with potential fishing regulations would be minimized in areas of conservation priority and so incidental capture of seabirds could be factored in (hereafter fishing costs). We assumed that economic costs and bycatch impacts would be proportional to fishing effort, so we used spatially explicit measurements of fishing effort as a proxy for fishing costs. Information on fishing pressure was provided by the Global Fishing Watch Data (<http://globalfishingwatch.org/>). These data are based on near real-time monitoring of global fishing activity and are used to compute the probability that a vessel is fishing in a particular area based on AIS tracking data (McCauley et al. 2016). Daily data were available for 2012–2017 in 8

Table 1. Protection target of seabird species breeding on the coast of Argentinean Patagonian under Birdlife International's Important Bird Area (IBAs) and 2 other protection scenarios modeled with the decision-support tool Marxan.^a

Species and feeding location	IUCN status ^b	Population size ^c	Target (% of total population protected) ^d		
			Birdlife's IBAs	IBA scenario	Reference scenario
Offshore feeder					
Magellanic Penguin (<i>Spheniscus magellanicus</i>)	NT	1,927,760	26.2	30.0	44.2
Southern Giant Petrel (<i>Macronectes giganteus</i>)	LC	4624	46.7	46.1	50.5
Inshore feeders					
King Cormorant (<i>Phalacrocorax atriceps</i>)	LC	97,872	40.8	50.3	58.3
Rock Shag (<i>Phalacrocorax magellanicus</i>)	LC	15,072	54.6	61.9	66.6
Red-legged cormorant (<i>Phalacrocorax gaimardi</i>)	NT	2182	0	30.0	47.5
Neotropic Cormorant (<i>Phalacrocorax olivaceus</i>)	LC	3024	8.4	60.5	58.3
Kelp Gull (<i>Larus dominicanus</i>)	LC	138,482	39.9	56.1	63.9
Dolphin Gull (<i>Leucophaeus scoresbii</i>)	LC	1248	35.2	52.8	54.9
Olrog's Gull (<i>Larus atlanticus</i>)	NT	186	58.7	66.6	73.5
South American Tern (<i>Sterna birundinacea</i>)	LC	28,132	36.63	41.1	46.5
Sandwich Tern (<i>Thalasseus sandvicensis eurygnatha</i>)	LC	4272	83.8	85.6	87.4
Royal Tern (<i>Thalasseus maximus</i>)	LC	1542	80.9	81.1	82.1
Brown Skua (<i>Stercorarius antarcticus</i>)	LC	974	80.2	87.3	89.6
Chilean Skua (<i>Stercorarius chilensis</i>)	LC	86	2.08	62.8	49.2

^a Considers the projected at-sea distribution model of seabirds and the economic costs of potential regulations in fishing practices.

^b International Union for the Conservation of Nature (IUCN) category abbreviations: LC, least concern; NT, near threatened.

^c Yorio et al. 1998.

^d Scenarios, Birdlife International Important Bird Areas (IBAs), candidate areas in the Patagonia; IBAs scenario, where IBAs were fixed to ensure they would be included in the resulting reserve network; reference scenario, combines solution obtained with Marxan with IBAs added subsequently to this solution obtained. Species conservation targets per area obtained in each scenario are given in Supporting Information.

× 8 km resolution raster format. For overall assessment of the spatial distribution of fishing pressure, we added daily information for the austral summer (January–March) (Fig. 1b) and the entire time frame (2012–2017 austral summers).

Reserve Network Design

We used the projected at-sea distribution of the breeding seabird community, IBAs in the study area (to include the distribution of other, mostly pelagic, seabird species occurring in Argentinean waters), and fishing costs (included as a penalty factor) to identify priority areas for conservation of the 14 seabird species breeding along the Patagonian coast. We applied the most commonly used free decision-support tool for conservation reserve network design, Marxan (version 1.8.10; <http://www.uq.edu.au/marxan/>) (Ball et al. 2009). We divided our study area into 11,858, 8 × 8 km planning units (similar resolution to that for fishing and environmental

data used) distributed within the continental shelf (200-m depth contour) where these central-place foragers typically occur during the breeding season.

Species-specific conservation targets (percentage of the total population to be protected) were predefined according to the seabirds' International Union for the Conservation of Nature (IUCN) status and based on aims of the IUCN World Parks Congress (2003) to ensure that at least 20% of the abundance of each species was represented in the resulting reserve network (Morfin et al. 2016). Accordingly, we defined population-protection targets for Patagonian colonies censused from 1993 to 1995 (Yorio et al. 1998) as 20% for species of least concern and 30% for near-threatened species (including *L. atlanticus*, *Phalacrocorax gaimardi*, and *S. magellanicus*).

Two different Marxan scenarios were considered depending on the way BirdLife marine IBAs for pelagic species were included in the analyses. In the first scenario (hereafter, IBAs scenario), IBAs were fixed

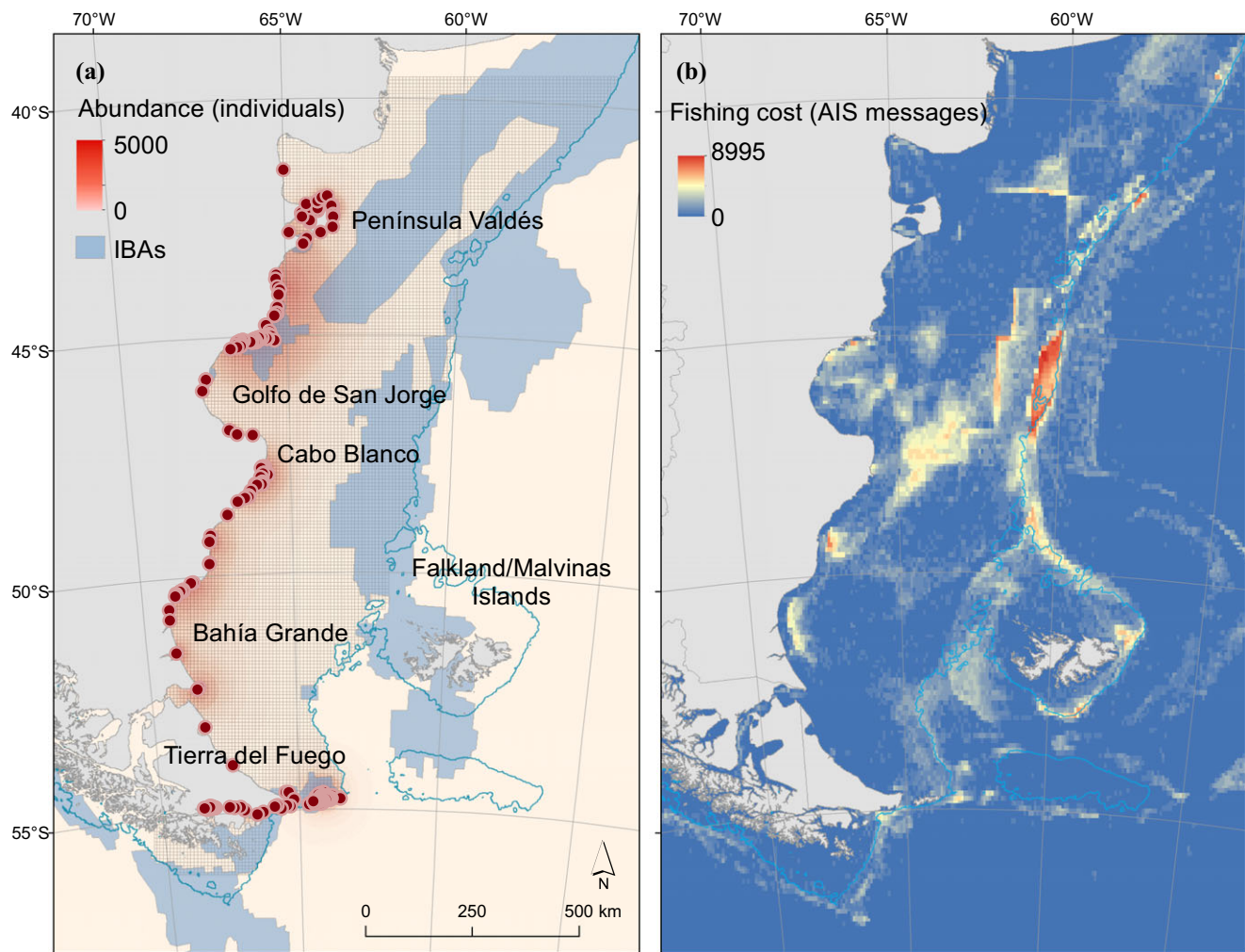


Figure 1. The (a) projected distribution of the Patagonian seabird breeding community based on breeding-pair colony censuses (shading, the darker the tone, the greater the number of individuals; points, seabird colonies surveyed; shaded marine areas, Birdlife International's Marine Important Bird and Biodiversity Areas [<https://maps.birdlife.org/marineIBAs/default.html>]; gridded areas, distribution of 8×8 km planning units in the study area) and (b) fishing-vessel locations determined by AIS messages used as fishing costs (marine line, continental shelf break, 200-m isobaths). These data were used as inputs in the reserve-network analysis with the decision-support tool Marxan.

within Marxan procedure to ensure they would be included in the resulting reserve network. In the second scenario (reference scenario), we first considered only projected at-set distribution areas of breeding seabirds and then marine IBAs were subsequently added to the reserve network obtained.

To accomplish conservation targets and keep some level of clumping in proposed areas for conservation (Ardrón et al. 2010), we calibrated species penalty factor (SPF) and boundary length modifier (BLM) with the free software Zonae Cogito. Calibration results were plotted as a trade-off curve between BLM and boundary length of the reserve system and SPF and missing values (Ardrón et al. 2010; Segán et al. 2011) (details given in Supporting Information). We conducted 1 million iterations and

repeated the procedure for 500 runs. For visual comparison, we displayed the best solution (the most efficient reserve configuration that incurred the lowest fishing costs and met biodiversity targets) and frequency solution (how often a given planning unit was selected in the final reserve system) among the 500 runs. Output reserves were the compact areas in the best solution.

Connectivity Analyses

We used information on marine currents in the study area to identify water masses that were highly connected to our priority areas for seabird conservation through water inflow (hereafter, connectivity areas). Monthly data on water speed and direction in the upper layers

(first 14 m) were retrieved from the global ocean physics reanalysis models (Glorys) 2 (version 3 1993–2012; www.myocean.eu, Copernicus: Marine environment monitoring service; 0.25° horizontal resolution). Mean water speed and direction for the 1993–2012 breeding months (September–February) were computed. Following Afán et al. (2015), we calculated the absolute angle difference (0–360°) between current azimuths and the direction of each pixel to each priority area. Angle differences were subsequently transformed into a measure of resistance by using a quadratic function that assigned minimum resistance values to angles pointing to priority areas. Resistance values gradually increased as the angle pointed increasingly away from priority areas. The final measure of resistance was obtained by multiplying the results by water speed; resistance values were highest for those locations where currents flowed the fastest away from (180°) priority areas.

Resulting resistance layers (1 layer per priority area) were incorporated in a connectivity-modeling approach in which circuit theory was used to estimate the cumulative probabilities of water masses flowing from source nodes (grid of 0.25° over the entire study area) to each priority area derived from Marxan (Supporting Information). Connectivity outputs for each priority area were added to a final connectivity map. Connectivity areas were delimited by considering those water masses enclosing the 98th quantile values for the connectivity map. Analyses were performed in Circuitscape software (McRae et al. 2008).

Marine Productivity and Pollution

We explored the distribution of marine productivity patterns and human impacts in the resulting connectivity areas. We used the longest time series of chlorophyll-a concentration (CHL) (1998–2017) to spatially and temporally evaluate the predictability of productive areas over time because CHL likely affects productivity patterns. Data were obtained from SeaWiFS and MODIS sensors (spatial resolution 0.083°, 9 km) for the austral summer seasonal composites (NASA Ocean Color Service [<http://oceancolor.gsfc.nasa.gov/>]). We defined highly productive pixels for each summer season as those with CHL values within the upper quartile (75th percentile) (Afán et al. 2014). Following Louzao et al. (2012), the persistence of ocean features was quantified by counting the number of years in which each pixel was assigned to the highly productive category (range 0–20 years corresponding to the CHL time series).

Land-based pollution is a minor human impact in our study area (Halpern et al. 2015), and the occurrence of ships at a particular location was assumed as a reasonable proxy for risks of point-source pollution, such as dumping and oil spills (Halpern et al. 2008; Grémillet & Boulinier 2009). Two different layers were downloaded

from global data sets of anthropogenic drivers of ecological change (<https://knb.ecoinformatics.org/#view/https://doi.org/10.5063/F1S180FS>) (Halpern et al. 2015). The first layer showed the spatial distribution of shipping lanes based on vessel locations and the second layer provided spatially explicit estimates of pollution coming from commercial shipping and ports. These 2 layers were standardized to maximum values to make them comparable and then added to obtain a final estimate of potential risk of point-source pollution. All analyses were performed in R (R Core Team 2013) and ArcGIS 10.2 (ESRI, Redlands, California, U.S.A.).

Results

Spatial Distribution of Seabirds and Human Impacts

The projected at-sea distribution of breeding seabirds along the Patagonian shelf showed that the highest densities occur north of Golfo de San Jorge, where the most abundant species (Magellanic Penguin and Kelp Gull [*Larus dominicanus*]) coexist (Fig. 1a). The Magellanic Penguin and the Red-legged Cormorant (*P. gaimardi*) were distributed along a large latitudinal range, whereas the other species had more localized colony distributions along the coastline (Supporting Information).

Fishing pressure concentrated in a wide area in the middle shelf east of Cabo Blanco, along the coast near Bahía Grande, and on the outer shelf limit east of Golfo de San Jorge. The most intensive fishing occurred at the shelf limit in front of Golfo de San Jorge, in an area of 250 km long (Fig. 1b).

Reserve Network

The best solution provided by Marxan in the IBAs scenario (Fig. 2a) was similar to the frequency solution (Supporting Information), which confirmed the best solutions were similar among Marxan runs. Species conservation targets achieved in the IBAs scenario far exceeded the proposed 20% for species of least concern and 30% for near-threatened species (Table 1) and ranged from 41.1% to 87.3% of the total Patagonian population for species of least concern and from 30% to 66.6% for near-threatened species. The IBAs scenario was 280,000 km² (4495 planning units), covered up to 20 potential areas of conservation priority, and included large pelagic areas along the continental shelf break in northern and central offshore regions (Fig. 2a). Coastal areas near Península Valdés south of Cabo Blanco and Tierra del Fuego were also selected. These areas differed in their extent (13–160,000 km²) and contribution to conservation targets (Supporting Information).

The Marxan best solution and frequency solution were similar for the reference scenario (Fig. 2b &

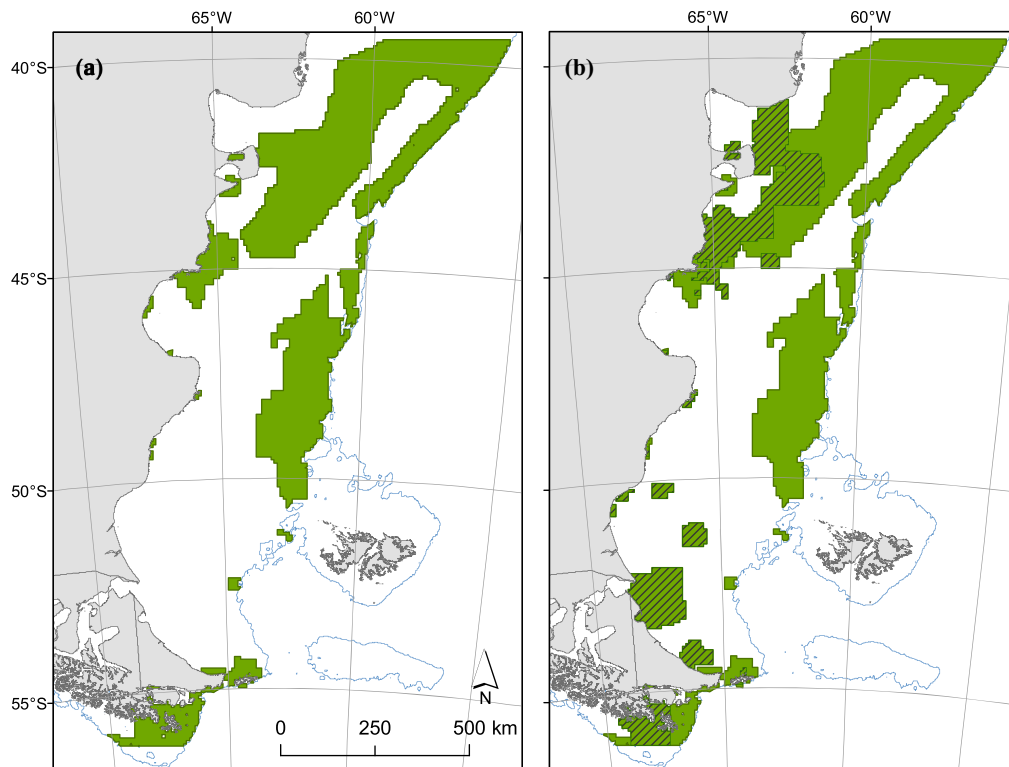


Figure 2. Proposed reserve networks along the Patagonian area identified with the decision-support tool Marxan based on the projected at-sea distribution model of seabirds and the economic costs of potential regulations on fishing practices (shaded areas, main priority conservation areas): (a) best solution for the IBAs (Birdlife International's Marine Important Bird and Biodiversity Areas) scenario, where candidate IBAs were fixed to ensure they would be included in the resulting reserve network and (b) solution for the reference scenario, which combines the best solution obtained with Marxan (striped area) with IBAs added to this best solution.

Supporting Information) and encompassed approximately 100,000 km² (1629 planning units) distributed in up to 20 potential areas of conservation priority that differed widely in their total extent (16–55,500 km²) (Fig. 2b). When IBAs were incorporated in this network, the total area preserved reached 330,000 km² (6281 planning units), 20% larger than for the IBA scenario, distributed in 23 potential areas. In addition to IBAs, priority areas were distributed along the Argentinean Patagonia coastline. The largest areas concentrated north of Golfo de San Jorge and south of Bahía Grande. Conservation targets were met for all species (Table 1) and ranged from 46.5% to 89.6% of the total Patagonian population for species of least concern and from 44.2% to 73.5% for near-threatened species. Priority areas greatly differed in their extent (13–194,000 km²) and contribution to conservation targets (Supporting Information).

Connectivity Areas

Three large connectivity areas distributed along the Argentinean Patagonia coastline appeared in both scenarios (Fig. 3 & Supporting Information). The largest extended from Cabo Blanco to Bahía Grande. Its total extent ranged

from 22,907 km² in the IBA scenario (area 1, Fig. 4a) to 17,123 km² in the reference scenario (area 4, Fig. 4b). Near Tierra del Fuego a highly connected area occurred (areas 2 and 5, Figs. 4a & 4b), the extent of which varied depending on the scenario (18,085 km² for IBA and 30,673 km² for reference). A somewhat smaller connectivity area occurred south of Península Valdés (area 3 in IBA scenario [3440 km²] and area 6 in reference [725 km²]). The connectivity areas differed in their productivity (Figs. 4c & 4d) and were subject to different levels of point source pollution (Figs. 4e & 4f). The central connectivity area (area 1 in IBAs scenario and 4 in reference) had the most productive pixels (Figs. 4c & 4d), whereas the northern connectivity area (areas 3 in IBAs scenario and 6 in reference) had the most shipping traffic and therefore had the highest risk of pollution (Figs. 4e & 4f).

Discussion

With our integrated approach to prioritization of marine areas for conservation, we identified 20 different priority areas that ranged in size from 280,000 to 330,000 km²,

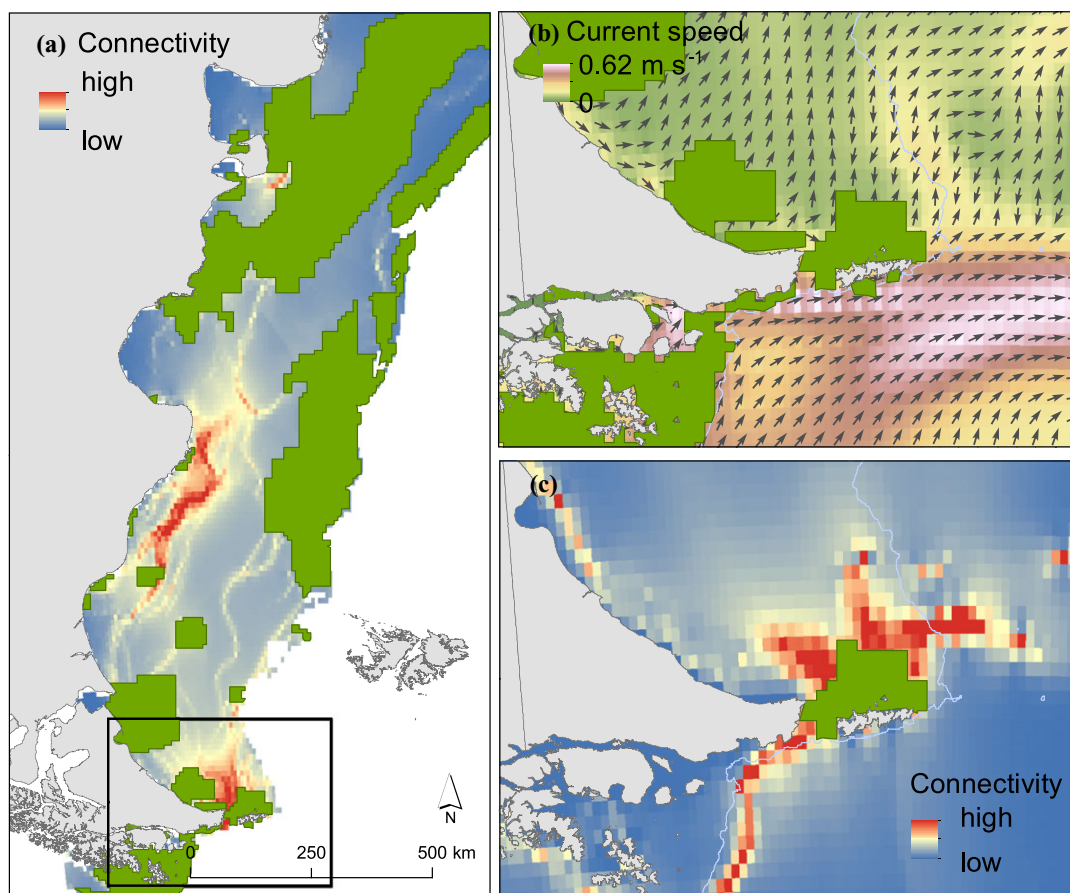


Figure 3. (a) Water masses connected with priority conservation areas (shaded) selected with Marxan analysis under Birdlife International's Marine Important Bird and Biodiversity Areas (IBA) scenario (candidate IBAs fixed to ensure they would be included in the resulting reserve network). Higher connectivity indicates a higher probability that these waters are included in the protected areas. Information on marine currents for the 1993–2012 breeding months (September–February) was used in a circuit-theory approach to identify main water masses connected to the reserve network through water inflow. Connectivity maps for each priority area were added to obtain final connectivity map (a). (b) Example from the southern Patagonian area of information on water speed and direction of marine currents used to identify those water masses connected to each priority area. (c) Connectivity map generated for a southern priority area in IBAs scenario.

depending on the scenario. We also identified 3 large areas that may affect the reserve network through water inflows. Our reserve network may benefit from enhanced marine productivity in these highly connected areas, but it may also be negatively affected by human impacts, such as dumping or oil spills.

Even though great efforts have been made to obtain spatial information on seabird movements (Catry et al. 2013; Ramos et al. 2016), tracking data are largely restricted to a few species and colonies (Kays et al. 2015), and it is improbable that data will be collected concurrently from all seabird colonies over large areas. By contrast, our approach coarsely extrapolates the distribution of large numbers of individuals of several species (14 in our case) distributed in a large area (here 3000 km). A source of uncertainty related to our method is a lack

of species and colony-specific information on foraging distributions. Further, our approach does not account for seabird foraging distances that may vary with colony size and sea area (Wakefield et al. 2017). Finally, individuals breeding outside our study area but feeding on the continental shelf were excluded from our at-sea projection but included in our reserve-network design because we considered IBAs. In the absence of specific guidelines on how to use these areas in the delimitation of marine protected areas, few researchers have included them in spatial planning prioritizations (but see McGowan et al. 2017).

Because seabirds breeding in Argentinean Patagonia distribute themselves preferentially in coastal waters near their colonies (Wilson et al. 2005; Boersma et al. 2009; Copello et al. 2011; Ramírez et al. 2014), offshore

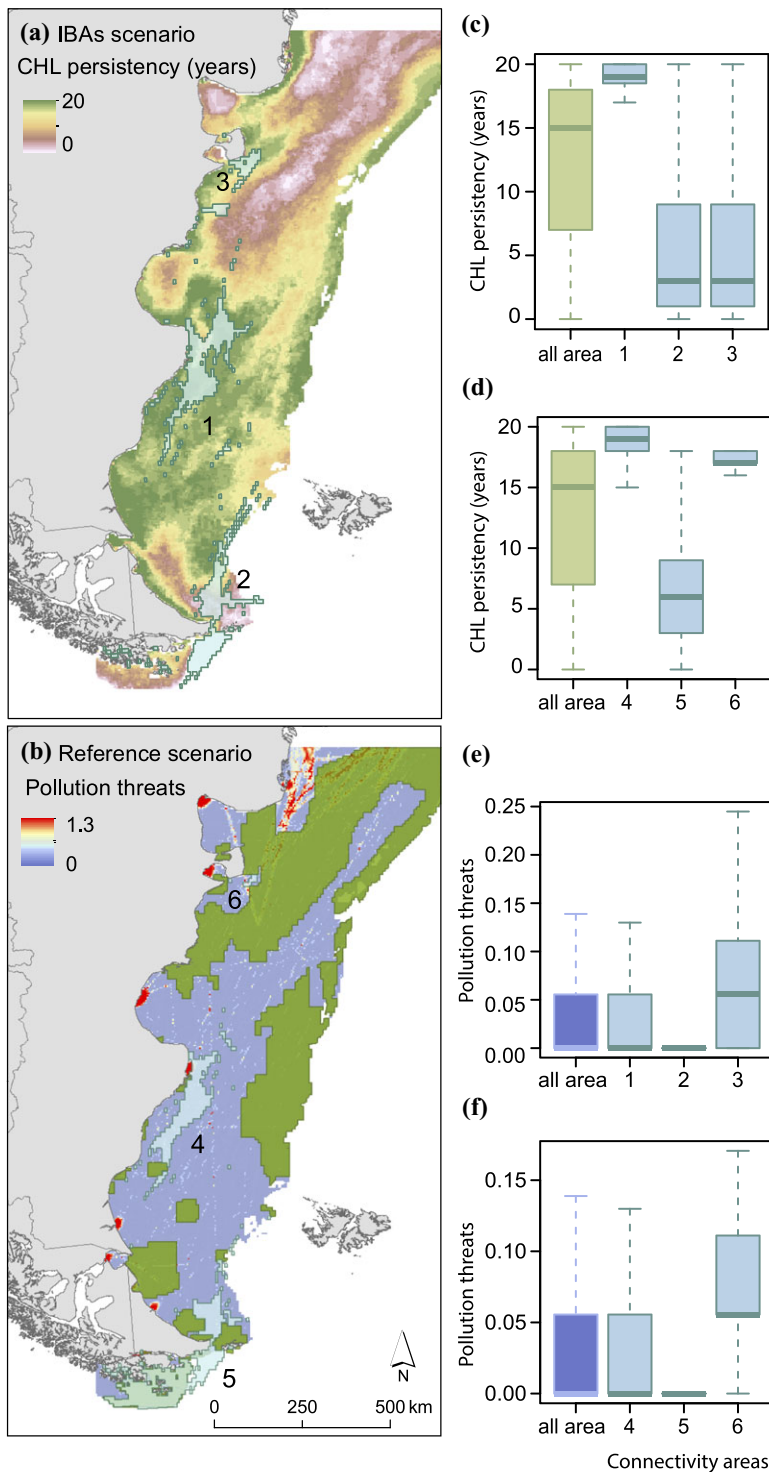


Figure 4. Connectivity areas (numbered areas), Patagonian shelf, where water masses with higher potential of reaching the priority areas (dark shaded areas in [b]) are enclosed for (a) Birdlife International Important Bird Areas (IBAs) scenario, where candidate IBAs were fixed to ensure they would be included in the resulting reserve network and (b) the reference scenario (which combines the best solution obtained with Marxan decision-support tool with IBAs added to the best solution obtained). These areas encompass waters with different conditions for chlorophyll-a (CHL) persistence (a, c, d) and pollution threats (b, e, f). Differences in (c, d) CHL and (e, f) pollution among the 3 main connectivity areas of each scenario (1–3, IBAs scenario; 4–6, reference scenario) (horizontal line within bars, median; bar ends, interquartile range; whiskers, farthest points that are not outliers). First bar in each graph shows values of CHL persistence (c, d) and pollution threats (e, f) for the entire study area.

IBAs alone do not fulfill conservation targets, and the reserve network from the IBAs scenario included additional coastal areas near Península Valdés, South of Cabo Blanco, and Tierra del Fuego. Many other coastal areas were also identified as priority areas when IBAs were excluded from the Marxan procedure but added a posteriori to the reserve network output (reference scenario). This was particularly true for the southern regions

in Bahía Grande, where potential priority areas for seabird conservation were under-represented in the IBAs scenario. Conservation targets for breeding seabirds in the reference scenario exceeded those in the IBAs scenario (particularly for near-threatened species) and included relevant marine protected areas of the southernmost populations. However, the reference scenario reserve network was also costlier than the IBAs scenario in terms

of total surface to be protected (1.2 times larger). Thus, the IBAs scenario maximized the efficiency in which limited economic resources can be allocated to create more meaningful and sustainable reserve networks (Beyer et al. 2016).

We assumed that economic costs of regulation in fishing practices within priority areas for seabird conservation would be proportional to the intensity of fisheries operating in these waters. However, there are ways to mitigate fisheries' impact on seabirds while minimizing the costs of fishing. For instance, a bottom-trawl fishery may not have adverse effects on surface-feeding birds, such as gulls or terns (it may, in fact, have a positive effect in terms of freely available discards [Tasker et al. 2000; Bicknell et al. 2013]), so banning bottom trawling in an area important for gulls and terns would not be necessary and could be counterproductive (Bicknell et al. 2013; Ramírez et al. 2015). Bycatch-mitigation measures could also be effective (Melvin et al. 2014) and would minimize fishing costs. Accurate knowledge of the ecological interactions between seabirds and fisheries would be desirable for implementing species and fishing-gear-specific mitigation measures to ensure seabird conservation and the sustainability of fisheries (Grémillet & Boulinier 2009; Bertrand et al. 2012).

Water movements through oceanic currents may also be relevant for the conservation of marine biodiversity in general and seabirds in particular (Scales et al. 2014; Afán et al. 2015). However, ocean dynamism and marine currents have been largely neglected in reserve design. Our approach is novel because it includes spatially explicit information on ocean currents to identify water masses likely to influence a reserve network through water inflows. Regardless of the scenario considered, we identified 3 highly connected areas along the Argentinean Patagonia coastline. Marine productivity patterns in these highly connected areas may benefit our reserve network by enhancing food availability through organism movement and biomass drift (Afán et al. 2015), whereas perturbations such as dumping and pollution events may expose protected areas to external threats (Halpern et al. 2008; Ruoppolo et al. 2012; Marinho et al. 2013). Our spatially explicit assessments indicated that marine productivity (as indicated by CHL values) in the central connectivity area was high and persistent over the years. Restrictions on fishing in this area may contribute to the functionality and protection of priority areas for seabird conservation by favoring food availability. In contrast, the connectivity area most subjected to potential risks of point-source pollution was in the northern region, near Península Valdés. Shipping restrictions or further control of shipping traffic in this area could minimize pollution risks and contribute to early detection of potential threats (i.e., oil spills that would likely reach priority areas for seabird conservation through marine currents, Johansen et al. 2003; Wang et al. 2005).

Our adaptive framework addresses quantitative conservation objectives, qualitative design criteria, and the dynamism and quality of the oceanographic habitat. In particular, it allows for the integration of an entire seabird community in the early stages of conservation planning. The resulting reserve network can be enhanced with additional information on the distribution of other animal groups (e.g., marine mammals or large predatory fish, Sund et al. 1981; Fiedler et al. 1998; Hunt Jr et al. 1998) or seabird breeding colonies (e.g., those in nearby areas such as the Falkland and Malvinas Islands) or with finer-scale information on seabird distribution or updated data on colony sizes. However, our proposed reserve network could be meaningless outside of the breeding season, when individuals are no longer linked to their breeding sites and disperse over broader regions (Phillips et al. 2006; Phillips et al. 2007). Complete protection of these seabird populations would benefit from information on their nonbreeding distribution and from dynamic reserve networks encompassing marine regions relevant for seabirds throughout their life cycle.

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Supporting Information

Probability distribution plots for inshore and offshore feeders (Appendix S1), projected at-sea distribution figures for the 14 species we considered (Appendix S2), calibration of SPF and BLM (Appendix S3), tables of conservation targets held for each of the species in the IBAs scenario (Appendix S4) and the reference scenario (Appendix S5), figures and supplementary methods of connectivity areas (Appendix S6), and figures of best and frequency solutions for the final proposed reserve network (Appendix S7) are available online. The authors are solely responsible for the content and functionality of these materials. Queries (other than absence of the material) should be directed to the corresponding author.

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